

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor.

$$[0 \quad 0 \quad Q_z]$$

$$Q_z = \mathbb{Q}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{xx} & 0 \\ 0 & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)}$$

$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & Q_{xy} & 0 \\ -Q_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xy} = \mathbb{G}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} Q_{xxx} & Q_{xxy} & Q_{xxz} \\ -Q_{xxx} & -Q_{xxy} & Q_{xxz} \\ 0 & 0 & Q_{zzz} \\ 0 & Q_{xxz} & 0 \\ Q_{xxz} & 0 & 0 \\ Q_{xxy} & -Q_{xxx} & 0 \end{bmatrix}$$

$$Q_{xxx} = \mathbb{Q}_{3a}^{(1)}$$

$$Q_{xxy} = \mathbb{Q}_{3b}^{(1)}$$

$$Q_{xxz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)}$$

$$Q_{zzz} = 3\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{xxz} \\ 0 & 0 & 0 \\ Q_{yzx} & -\frac{Q_{xxz}}{2} & 0 \\ -\frac{Q_{xxz}}{2} & -Q_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = 2\mathbb{Q}_z^{(2)}$$

$$Q_{yzx} = 3\mathbb{G}_u^{(1)}$$

$$\frac{Q_{xxz}}{2} = \mathbb{Q}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} Q_{yzx} & Q_{yzy} & 0 \\ -Q_{yzy} & Q_{yzx} & 0 \\ 0 & 0 & Q_{xyz} \end{bmatrix}$$

$$Q_{yzx} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(2)}$$

$$Q_{yzy} = \mathbb{Q}_z^{(3)}$$

$$Q_{xyz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & -Q_{xxyz} & -Q_{xxzx} & 0 \\ Q_{xxzz} & Q_{xxzz} & Q_{zzzz} & 0 & 0 & 0 \\ Q_{xxyz} & -Q_{xxyz} & 0 & Q_{yzyz} & 0 & -Q_{xxzx} \\ Q_{xxzx} & -Q_{xxzx} & 0 & 0 & Q_{yzyz} & Q_{xxyz} \\ 0 & 0 & 0 & -Q_{xxzx} & Q_{xxyz} & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$

$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)}$$

$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)}$$

$$Q_{xxyz} = \mathbb{Q}_{4a}^{(1)}$$

$$Q_{xxzx} = \mathbb{Q}_{4b}^{(1)}$$

$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxyy} \\ 0 & 0 & Q_{xxzz} & -Q_{xxyz} & -Q_{xxzx} & -Q_{xxyy} \\ -Q_{xxzz} & -Q_{xxzz} & 0 & 0 & 0 & 0 \\ -Q_{xxyz} & Q_{xxyz} & 0 & 0 & Q_{yzzz} & Q_{xxzx} \\ -Q_{xxzx} & Q_{xxzx} & 0 & -Q_{yzzz} & 0 & -Q_{xxyz} \\ -Q_{xxyy} & Q_{xxyy} & 0 & -Q_{xxzx} & Q_{xxyz} & 0 \end{bmatrix}$$

$$Q_{xxzz} = 3\mathbb{Q}_u^{(3)}$$

$$Q_{xxyz} = 2\mathbb{G}_{3a}^{(1)}$$

$$Q_{xxzx} = -2\mathbb{G}_{3b}^{(1)}$$

$$Q_{xxyy} = 2\mathbb{G}_z^{(1)} - 2\mathbb{G}_z^{\alpha(1)}$$

$$Q_{yzzx} = -\mathbb{G}_z^{(1)} - 4\mathbb{G}_z^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{yzyz} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)}$$

$$Q_{xyxy} = \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & Q_{yzzx} & 0 \\ -Q_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzzx} = -\mathbb{G}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} Q_{xxyz} & Q_{xxzx} & Q_{xxxy} \\ -Q_{xxyz} & -Q_{xxzx} & Q_{xxxy} \\ 0 & 0 & Q_{zzxy} \\ Q_{yzyz} & Q_{yzzx} & 0 \\ Q_{yzzx} & -Q_{yzyz} & 0 \\ Q_{xxzx} & -Q_{xxyz} & 0 \end{bmatrix}$$

$$Q_{xxyz} = \mathbb{G}_{3a}^{(2)}$$

$$Q_{xxzx} = \mathbb{G}_{3b}^{(2)}$$

$$Q_{xxxy} = \mathbb{G}_z^{(3)} - \mathbb{G}_z^{\alpha(2)}$$

$$Q_{zzxy} = \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{(4)} + 2\mathbb{G}_z^{\alpha(2)}$$

$$Q_{yzyz} = -3\mathbb{Q}_u^{(5)}$$

$$Q_{yzzx} = \mathbb{G}_z^{(4)} - \mathbb{G}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} Q_{yzxx} & -Q_{yzxx} & 0 & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{yzxy} & -Q_{yzxy} & 0 & Q_{yzzx} & -Q_{yzyz} & -Q_{yzxx} \\ Q_{xyxx} & Q_{xyxx} & Q_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzxx} = \mathbb{G}_{3a}^{(3)}$$

$$Q_{yzyz} = -3\mathbb{Q}_u^{(6)}$$

$$Q_{yzzx} = \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)}$$

$$Q_{yzxy} = \mathbb{G}_{3b}^{(3)}$$

$$Q_{xyxx} = \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)}$$

$$Q_{xyzz} = \mathbb{G}_z^{(5)} + 2\mathbb{G}_z^{(6)} + 2\mathbb{G}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxxy} & Q_{xxzy} & Q_{xxxz} & Q_{xxyx} \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & -Q_{xxyz} & -Q_{xxzx} & -Q_{xxyx} & -Q_{xxzy} & -Q_{xxxz} & -Q_{xxyy} \\ Q_{zzxx} & Q_{zzxx} & Q_{zzzz} & 0 & 0 & Q_{zzxy} & 0 & 0 & -Q_{zzxy} \\ Q_{yzxx} & -Q_{yzxx} & 0 & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} & Q_{yzzy} & Q_{yzxz} & Q_{yzxy} \\ -Q_{yzxy} & Q_{yzxy} & 0 & -Q_{yzxz} & Q_{yzzy} & Q_{yzxx} & -Q_{yzzx} & Q_{yzyz} & Q_{yzxx} \\ -\frac{Q_{xxxxy}}{2} - \frac{Q_{xxyx}}{2} & \frac{Q_{xxxxy}}{2} + \frac{Q_{xxyx}}{2} & 0 & -Q_{xxxz} & Q_{xxzy} & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} & -Q_{xxxz} & Q_{xxzy} & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$

$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)}$$

$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)}$$

$$Q_{xxyz} = 2\mathbb{G}_{3a}^{(1)} + \mathbb{G}_{3a}^{(2)} + \mathbb{Q}_{4a}^{(1)}$$

$$Q_{xxzx} = -2\mathbb{G}_{3b}^{(1)} + \mathbb{G}_{3b}^{(2)} + \mathbb{Q}_{4b}^{(1)}$$

$$Q_{xxxy} = 2\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)}$$

$$Q_{xxzy} = 2\mathbb{G}_{3a}^{(1)} - \mathbb{G}_{3a}^{(2)} + \mathbb{Q}_{4a}^{(1)}$$

$$Q_{xxxz} = -2\mathbb{G}_{3b}^{(1)} - \mathbb{G}_{3b}^{(2)} + \mathbb{Q}_{4b}^{(1)}$$

$$Q_{xxyx} = 2\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)}$$

$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)}$$

$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{zzxy} = \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{(4)} + 2\mathbb{G}_z^{\alpha(2)}$$

$$Q_{yzxx} = -2\mathbb{G}_{3a}^{(1)} + \mathbb{Q}_{4a}^{(1)}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)}$$

$$Q_{yzzx} = -\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(4)} - 4\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)}$$

$$Q_{yzxy} = -2\mathbb{G}_{3b}^{(1)} - \mathbb{Q}_{4b}^{(1)}$$

$$Q_{yzzx} = \mathbb{Q}_0^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)}$$

$$Q_{yzxz} = -\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(4)} - 4\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} Q_{yzxx} & -Q_{yzxx} & 0 & Q_{yzyz} & Q_{yzxz} & Q_{yzxy} & Q_{yzzx} & Q_{yzxz} & Q_{yzxy} \\ Q_{yzxy} & -Q_{yzxy} & 0 & Q_{yzxz} & -Q_{yzzx} & -Q_{yzxx} & Q_{yzzx} & -Q_{yzyz} & -Q_{yzxx} \\ Q_{xyxx} & Q_{xyxx} & Q_{xyzz} & 0 & 0 & Q_{xyxy} & 0 & 0 & -Q_{xyxy} \end{bmatrix}$$

$$Q_{yzxx} = \mathbb{G}_{3a}^{(3)}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)}$$

$$Q_{yzzx} = -\mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)}$$

$$Q_{yzxy} = \mathbb{G}_{3b}^{(3)}$$

$$Q_{yzzx} = -\mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)}$$

$$Q_{yzxz} = \mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)}$$

$$Q_{xyxx} = \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)}$$

$$Q_{xyzz} = \mathbb{G}_z^{(5)} + 2\mathbb{G}_z^{(6)} + 2\mathbb{G}_z^{\alpha(3)}$$

$$Q_{xyxy} = \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & 0 & & & Q_{xxxx} \\ 0 & Q_{xxxx} & & & Q_{yyyy} \\ 0 & 0 & & & Q_{zzzz} \\ -Q_{xxxx} & Q_{yyyy} & & & Q_{yyzz} \\ Q_{zzxx} & 0 & & & 0 \\ 0 & Q_{xyxy} & & & -Q_{yyyz} \\ 0 & Q_{yzyz} & & & 0 \\ Q_{xxxz} & -Q_{yyyz} & -\frac{Q_{xxxx}}{2} + \frac{3Q_{xyyy}}{2} + Q_{yyzz} - Q_{yzzx} + Q_{zzxx} \\ \frac{Q_{xxxx}}{2} - \frac{Q_{xyyy}}{2} & 0 & & & -Q_{xxxz} \\ -Q_{yyyz} & -Q_{xxxx} & & & 0 \end{bmatrix}$$

$$Q_{xxxx} = 3\mathbb{Q}_0^{(1)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$

$$Q_{xxxz} = \mathbb{Q}_{4b}^{(1)}$$

$$Q_{yyyz} = -\mathbb{Q}_{4a}^{(1)}$$

$$Q_{zzzz} = 3\mathbb{Q}_0^{(1)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{yyzz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)}$$

$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)}$$

$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)}$$

$$Q_{yzzz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)}$$